

Practice Quiz: Action

Name: __ Date: __

Part A: Multiple Choice

1. In Active Inference, action is best described as: A) Seeking reward and avoiding punishment B) Changing the world to make sensory inputs match the agent's predictions C) Random movement until something works D) Following predetermined motor programs
2. Expected Free Energy (EFE) evaluates: A) How much energy an action will consume B) How much future surprise an action is expected to produce C) The cost of an action in calories D) Whether an action is morally right
3. Pragmatic value refers to: A) How practical an action is B) The component of EFE that measures whether an action brings preferred outcomes C) The financial cost of an action D) How quickly an action can be performed
4. Epistemic value refers to: A) How much an action costs to learn B) The component of EFE that measures how much an action reduces uncertainty C) Academic knowledge D) The precision of sensory data
5. The exploration-exploitation dilemma is solved in Active Inference by: A) Always exploring B) Always exploiting C) EFE naturally balancing epistemic (explore) and pragmatic (exploit) value D) Flipping a coin
6. Motor control in Active Inference works by: A) The brain sending direct commands to muscles B) Muscles acting as reflexes that minimize prediction error between predicted and actual body position C) Conscious decision to move each muscle D) Spinal cord operating independently from the brain
7. Parkinson's disease difficulty with movement initiation can be understood as: A) Muscle weakness B) Impaired precision on action predictions due to dopamine loss C) Cognitive confusion D) Too much exploration

Part B: Short Answer

1. Compare how a traditional psychologist and an Active Inference researcher would explain why a person reaches for a glass of water. The traditional view uses desire/reward; the Active Inference view uses prediction. Explain both accounts.
2. You're at a buffet with 15 dishes you've never tried. Explain how EFE (epistemic and pragmatic value) would guide your sampling strategy as you try dishes one by one.
3. Explain why Active Inference predicts that newborn babies should be highly exploratory (trying everything, putting things in their mouths) while adults should be more exploitative (sticking with what works). Use the concepts of uncertainty and epistemic value.